

Research team, previous output and student training (max. 4 pages)

A. DESCRIPTION OF RESEARCH TEAM

The theme of public transportation and equity is multifaceted and requires an appropriate mix of skills and expertise. In response, this project gathers a group of researchers with complementary skills and perspectives. The research team is composed of one principal investigator, four co-investigators and one international collaborator, and includes scholars in the social sciences (Páez, Higgins, Ravensbergen, and Pereira), humanities (De Barros), and engineering (Páez and Mohamed). The principal investigator (Páez) and two of the co-investigators (Higgins and Ravensbergen) are alumni of the SSHRC-funded project Mobilizing Justice. One of the co-investigators (De Barros) is Director of McMaster's Centre for Human Rights and Restorative Justice, and another (Mohamed) is Director of McMaster Institute for Transportation and Logistics. Profiles and roles in the project are as follows.

Juanita DE BARROS (co-applicant, relative contribution to project 15%, 15% of personal research time) is Professor of History at McMaster University and the director of the Centre for Human Rights and Restorative Justice. She is an internationally-recognized historian whose work explores the intersection of race, gender, and class in the construction of public policy, including in colonized urban spaces. This domain of knowledge is key to the project, as it combines social and government policy in ways that affect racialized and genderized people in urban spaces. De Barros expertise in discourse analysis and in the field of human rights is central to the aims of this project, where a clear understanding of stakeholders about the grounds for justice in public services is essential. For this project, she will lend her expertise and vision through the co-supervision of trainees.

Christopher D. HIGGINS (co-applicant, relative contribution to project 15%, 15% of personal research time) is Associate Professor at the University of Toronto Scarborough in the Department of Human Geography and in the Department of Geography and Planning at the University of Toronto. He is among the world's top 2% of researchers in the Elsevier-Stanford rankings and received the 2025 University of Toronto Scarborough's New Scholar Award for exceptional research achievement and impact among early career faculty. He has experience working with public sector partners such as the City of Toronto, City of Hamilton, and Ontario Ministry of Transportation. His research uses geospatial analysis and accessibility measurement to examine how transportation systems shape access to housing, employment, and essential services, with particular attention to social equity. His work has been supported by SSHRC and NSERC, most recently through an NSERC Alliance Grant supporting the University of Toronto's tri-campus Mobility Network Consortium. For EquiTransit, he will lead the development of geospatial analytics, reproducible data pipelines, and interactive analytical tools.

Moataz MOHAMED (co-applicant, relative contribution to project 15%, 15% of personal research time) is associate Professor of Smart Systems and Transportation at McMaster University and Director of the McMaster Institute for Transportation and Logistics (MITL). He is the 2025 recipient of the Joseph Ip Distinguished Engineering Fellowship and is featured Elsevier-Stanford list of the world's top 2% of scientists. Mohamad is an experienced researcher, with more than 90 peer-reviewed articles and over \$6 million in secured research funding to his credit. He has trained 57 highly qualified personnel, including 11 PhD students, 12 master's students, and 7 postdoctoral fellows. Since 2018, he has been principal investigator on a City of Hamilton-funded program to reenvision the city's transit operator, Hamilton Street Railway (HSR). He is the principal developer of the AI-Transit Orchestrator, an \$1.4M NRCan-funded fleet-planning toolkit co-developed with Halifax Transit, Calgary Transit, Niagara Region Transit, and ETUQ. On this proposal, he will lead the team's work to integrate equity considerations in transit planning operations, translating multi-agency networks into a GTFS-based equity analytical model and equity-informed planning framework.

Antonio PÁEZ (principal applicant, relative contribution to project 30%, 80% of personal research time) is Professor of Transport Geography at McMaster University. He is a seasoned researcher who led the multi-institutional team that won the 2010 Meredith F. Burrill Award of the American Association of Geographers for exceptional work at the intersection of policy and basic research. He is also the recipient of the 2024 Edward L. Ullman Award of the American Association of Geographers for lifelong contributions to transportation geography. Páez has published widely, including over 170 peer-reviewed publications, three edited books, and a textbook. Páez is among the top 20 transportation researchers in the world whose careers started after 1999 according to Elsevier-Stanford's ranking, and is best known for his social science theory-informed quantitative research. This includes work on *normative accessibility* and *relative accessibility*, which aimed at implementing equity ideals in accessibility analysis. He is also well-versed in theoretical and qualitative approaches. He will direct the project and will participate in research and supervision of graduate students across the project.

Rafael H.M. PEREIRA (collaborator, relative contribution to project 10%, 5% of personal research time) is Director of Data Science at IPEA, Brazil's federal economics research institute. His contributions to transportation justice are largely responsible for shaping the field (27,32,66,67), while his contributions to numerous publicly available *R* and *Python* packages have been hugely influential in changing the practice of transportation research and planning. Open-source software such as {r5r} (57) and {r5py} (68) (which collectively have over a million downloads) are now routinely used by community planners, geographic researchers, and other members of the public for the purpose of simulating routing by different transportation modes. He has also worked on several packages for working with general transit feed specification data such as {gtfstools} (68). His role will be to provide advice on the theory and practice of transportation justice, and he will also lend his considerable expertise to the development of open-source tools.

Léa RAVENSBERGEN (co-applicant, relative contribution to project 15%, 15% of personal research time) is an early career researcher of equity in sustainable urban mobility. She is a mixed-methods researcher who has led multiple qualitative research projects focused on transportation users' experiences, including two that utilized a Codesign framework. In 2025, she led a series of Codesign focus groups with older adults on how best to design mobility reporting to encourage active lifestyles (currently under review in *Journal of Aging and Physical Activity*). In 2026, she worked with the Hamilton's transit operator Hamilton Street Railway (HSR) to Codesign how best to deliver paratransit as the City of Hamilton moves toward integrated transit. Ravensbergen will lead the Codesign component of the project.

B. DESCRIPTION OF PREVIOUS AND ONGOING RESULTS

Juanita DE BARROS's main field of research is into human rights and justice, with a focus on the political and social factors that shaped policies dealing with the intersection of health and the law in the past, their impact on the lived experience of women and children especially, and the longer-term impact of these policies. Her research projects include a sole-authored monograph on gender and reproductive health in the Caribbean (75), a forthcoming co-written article on urban vagrancy laws in late nineteenth-century Guyana, and a multidisciplinary co-edited essay collection (with McGill-Queens University Press). This latter project addresses child protection policies and discourse from the nineteenth to the early twenty-first centuries and the varied ways that these policies served adult institutional and political interests, often resulting in the forced labour and incarceration of children. This project is De Barros first foray into transportation-related research, and we believe that her historian's eye, in addition to her expertise in human rights and justice, will be an asset to the proposed work. Historical perspectives on

transportation remain rare despite their value (76,77) and collaboration between historians and transportation scholars is a unique aspect of the project.

Christopher D. HIGGINS has done research with conceptual, methodological, and applied outputs that are relevant to the proposed work. This includes the development of place-based/person-based (69), as well as competitive accessibility measures (43,70) that improve our understanding of who can reach destinations using the transportation network. His applied studies have incorporated equity into route-level transit performance measurement and examined disparities affecting the access of low-income residents, and essential workers to health and social services (38,71,72). Complementary research has assessed relationships among transit accessibility, residential location, and housing prices within individual metropolitan regions and across Canada (73). His ongoing research extends these approaches to the joint analysis of equity in accessibility and housing affordability. Further, Higgins brings a commitment to open, replicable, and reproducible research as evidenced through the development of the {pynogrid} package (74) and his work on a forthcoming open access book “Advanced Accessibility Analysis with R”. These skills mean that EquiTransit is well-grounded for developing theoretically-informed equity indicators, for conducting comparable analyses across Canadian metropolitan areas, and for communicating complex patterns through accessible analytical tools.

Moataz MOHAMED is an expert in developing and evaluating indicators of transit service performance and quality from both operator and rider perspectives (78–80). He has led qualitative and mixed-methods investigations into the needs of current and potential riders of transit, including grounded-theory studies of the barriers travelers face (81,82). As well, he has studied the satisfaction of transit users among population with distinct mobility needs (83). Mohamed is an expert in travel behaviour and preference heterogeneity across rider personas and generational cohorts (84–86), and in the development of accessibility and spatial-analysis measures for transportation (70,87,88). He has direct experience translating research into open, web-based decision-support tools for public agencies and practitioners, including work applying open-source data and machine learning to inform transportation decision-making (89). He has developed public-facing tools for the Ontario Ministry of Transportation. Currently, Mohamed is Director of the McMaster Institute for Transportation & Logistics and PI/Co-PI on a portfolio of NSERC-, NRCan-, and municipally funded projects on electric and zero-emission transit system planning, resilience, and equitable transit service design.

Antonio PÁEZ (Principal Investigator) has extensive experience with the study of the transportation challenges faced by transport-poor populations, including older adults (90–93), single parents (94,95), and children (96–98). His longstanding interest in the implications of social science theory and policy on methodology is illustrated by his research on normative accessibility (99) and relative accessibility (100), work initially motivated by the policy needs of Human Resources and Social Development Canada (an agency of Canada’s federal government). His interests now encompass broader issues of equity and justice in transportation (64,71,101). As a member of the SSHRC-funded project Mobilizing Justice, his research led to novel tools for the analysis of accessibility (43,70,102), clearer theoretical foundations for developing standards of equity (45), and the role of human dignity in transportation (103). Páez is an expert in the development and application of spatial analysis tools with a focus on transportation (87,94,104) and the development of web-based tools for public use (105,106), and he is also fluent in the use of qualitative methods (82,107). His breadth of knowledge and expertise is an asset for this project, in his role as PI and transversal conduit between the various work packages.

Léa RAVENSBERGEN has researched the transit experiences of equity-seeking groups (83,108,109) and the systemic barriers transit agencies face in planning just systems (108,110,111). Her expertise is as a qualitative methods researcher (109,112–114) including mobile ethnography (115,116). She has conducted Codesign research in which older adults codeveloped a mobility reporting tool (117) and has expertise in community-based research (118). Her work includes collaborations with all levels

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of governments in various provinces, transit agencies across the country, and non-for-profit organizations. She is part of a pan-Canadian team that was granted a \$.16 million NSERC Alliance grant to research data-driven tools for planning and operating gender-inclusive public transit systems in Canada. In this role, she is working in partnership with 14 Canadian transit agencies to better integrate and implement gender-sensitive data collection practices. Since 2023 she has worked with the Hamilton Street Railway (HSR), the public transit agency of the City of Hamilton (108,119,120). She is well positioned to lead to the Codesign element of this research, and has the relationships required to ensure a successful process.

C. DESCRIPTION OF TRAINING STRATEGIES

Our strategy for training highly qualified personnel is based on extensive involvement of the students, well-defined responsibilities for the integration of graduate students into all aspects of the project, and close supervision by a team of interdisciplinary supervisors (see Table MT.1).

Table MT.1. Supervisory Matrix

			Year	Main Supervisor	Supervisory Committee
WP.1 Theoretically-grounded equity indicators	Post-Doctoral Fellow 1	PD1	1	Páez	De Barros, Pereira
	Post-Doctoral Fellow 2	PD2	5	Páez	De Barros, Pereira
WP.2 Codesign	Ph.D. Student 1	PhD2	1 - 4	Ravensbergen	Páez, De Barros
WP.3 Geospatial data analytics	Ph.D. Student 2	PhD1	2 - 5	Higgins	Páez, Pereira
WP.4 Equity-informed transit planning and evaluation	Master Student 1	M1	1 - 2	Mohamed	Páez, Higgins
	Master Student 2	M2	3 - 4	Mohamed	Páez, Higgins

PD1 and PD2 will work with Páez and De Barros across work packages. PD1 will be part of WP.1 and will interact mainly with the geospatial data analytics package (WP.3) on the data analytics needed for the codesign work package (WP.2) to communicate with stakeholders. In the latter part of the project, PD2 will liaise primarily with the codesign package (WP.2) to assess the usability and performance (user experience) of EquiTransit tools. PhD and Master's students will maintain a tighter focus. PhD1 will be on the codesign package (WP.2) and PhD2 will focus on geospatial technologies and the development of open-source tools (WP.3). M1 and M2 will participate in the development of equity-based assessment and evaluation, beginning with a base tool that will incorporate equity-informed indicators after they have been validated as part of the codesign package (see **Table T1**). Through the project, all trainees will develop a variety of soft and transferable skills (see **Table MT.2**) through a combination of close supervision and regular discussion, and enrolment in appropriate graduate courses [e.g., reproducible research; (121)].

Table MT.2. Skills Matrix

	Year	KS	QL	QT	PP	DL	PC	DM	RE	CCE	TW	IIC	TM	BM
PD 1	1													
PD 2	5													
PhD 1	1 - 4													
M 1	1 - 2													
M 2	4 - 5													
PhD 2	2 - 5													

Skills (Key): **KS:** Knowledge Synthesis; **QL:** Qualitative research methods; **QT:** Quantitative research methods; **PP:** Publication and presentations (communication skills); **DL:** Digital literacy; **PC:** Programming and coding; **DM:** Data management and analysis; **RE:** research ethics; **CCE:** Consultation and community engagement; **TW:** Teamwork; **IIC:** International and intersectoral conference; **TM:** Time management; **BM:** Budget management.